

Quantum Queue Utility

Quantum HPC integration tool

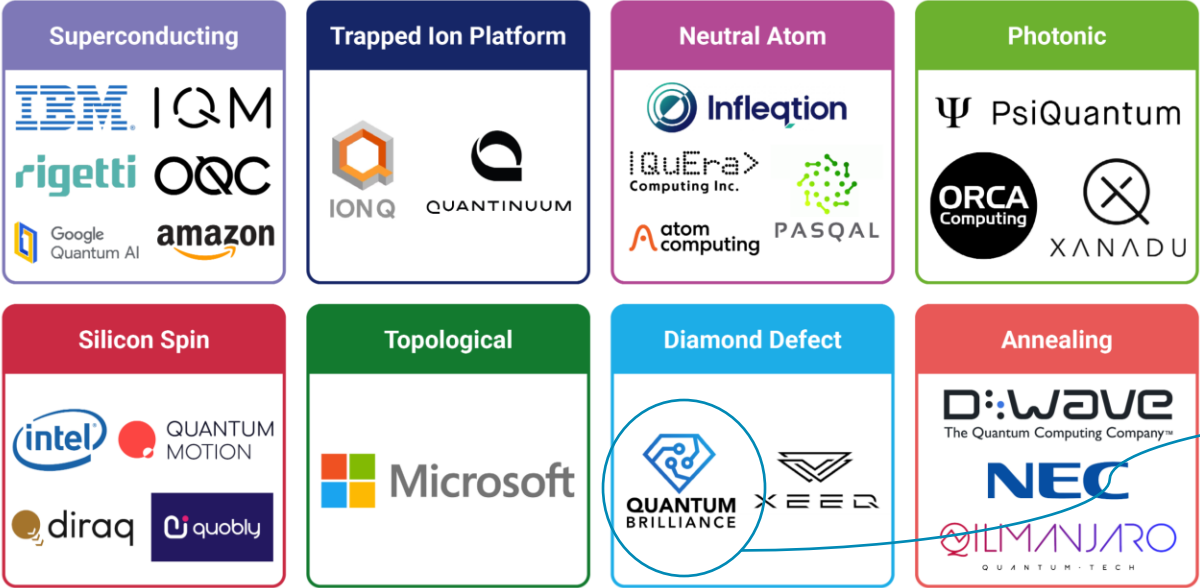


Christian Ortiz

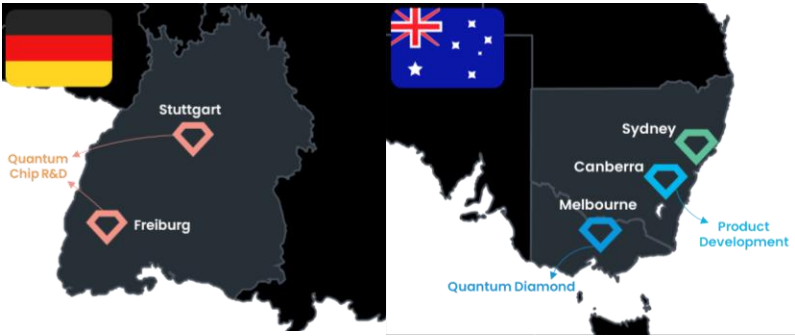
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Quantum Brilliance Context

Eight leading approaches to build a commercial quantum computer



Reference: <https://www.idtechex.com/de/research-article/how-the-quantum-physics-nobel-links-to-a-future-us-21b-market/33858>



Quantum utility at room temperature



Research areas / Projects



- quantum sensing
- pulse level access
- multi-NV algorithms
- quantum accelerators
- quantum development kits
- mobile quantum computing

QB Partners & Customers



QB Investors



Quantum Brilliance Roadmap

Supercomputers and quantum machines are now officially talking to each other

Damien Black, Senior Journalist

cybernews®

Computer researchers claim to have run the first successful experiment that proves quantum computers can work in tandem with their classical counterparts, using algorithms to communicate.

"The first algorithm was run between an on-premises quantum computer and supercomputer, validating [that] quantum and classical computing systems can communicate to solve complex challenges," said Pawsey Supercomputing Center and Quantum Brilliance in a press release shared with Cybernews.

The Pawsey center ran what it described as "its first quantum algorithm" on a room-temperature diamond-based system installed at a facility run by its partner Quantum Brilliance in Perth, Australia, on April 18.



Liz Bridge (She/Her) • 1st
Technical Staff Officer in the CTO Office @ Quantum Brilliance | Oak...

World first: Quantum computing in the Australian bushland.

LinkedIn News Australia
1,546,552 followers
2w •

An Australian tech startup has run its mini quantum computers on an electric car battery in the bush, in what it claims to be world-first.

...see more

My car ran a quantum computer
Simon Gemmell on LinkedIn • 3 min read

TIME BEST INVENTIONS 2025

QUANTUM BRILLIANCE



2022

2023

2025

Coming in 2028

World's 1st
Integration inside
a Supercomputer

World's 1st
Mobile
Quantum Computing

World's 1st
Quantum Computer Cluster

World's 1st
GPU-sized Mobile Quantum
computer



Quantum Queue Utility is currently being used in 3 HPC centers



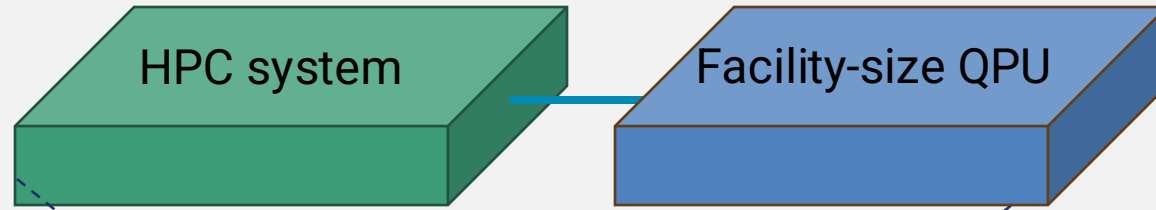
Quantum-HPC integration

What is next?

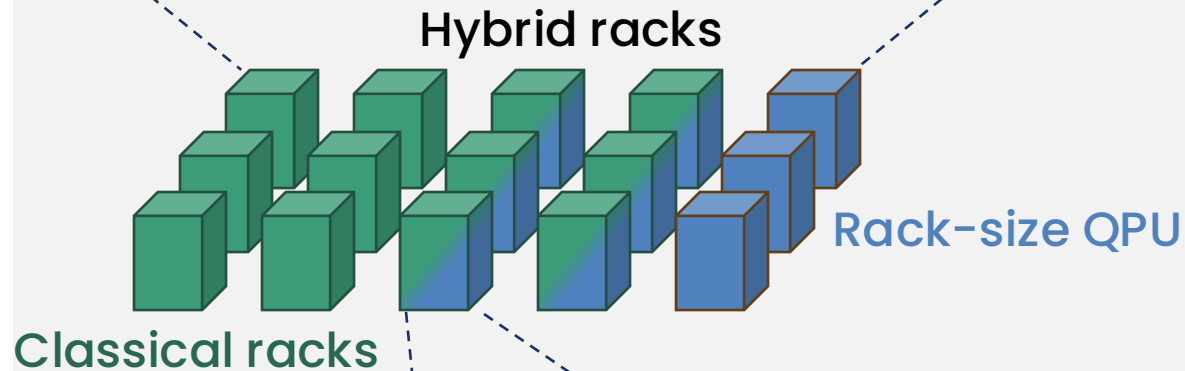
Levels of Q-HPC integration

Quantum
Classical

Level 1

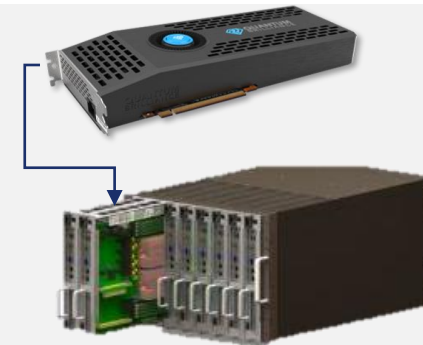
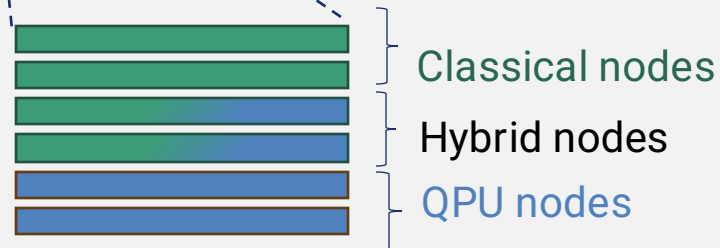


Level 2



- High speed infiniband / ethernet. High level protocols
- Ab initio chemistry, optimization applications
- EM isolation, vibrational isolation, cryogenics

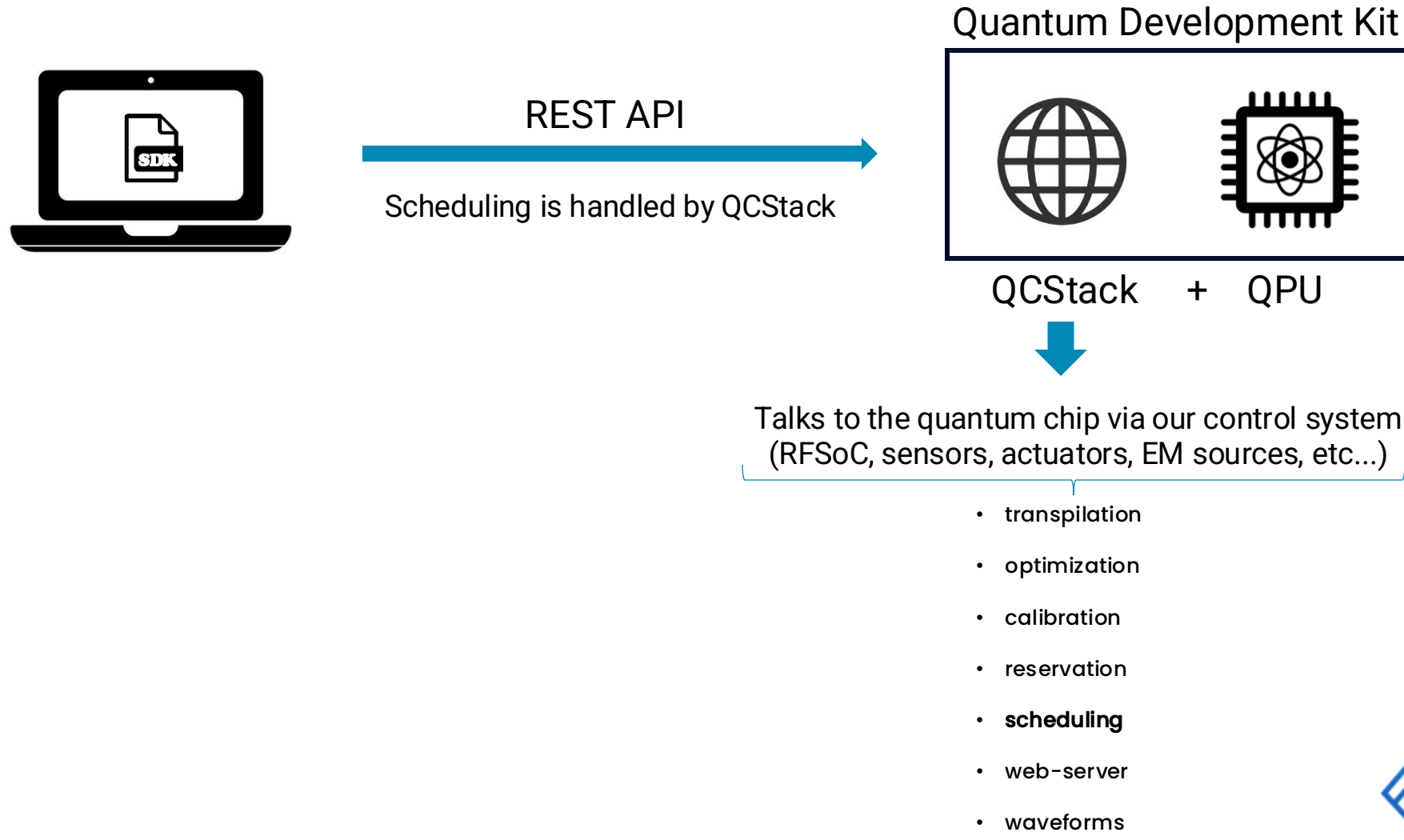
Level 3



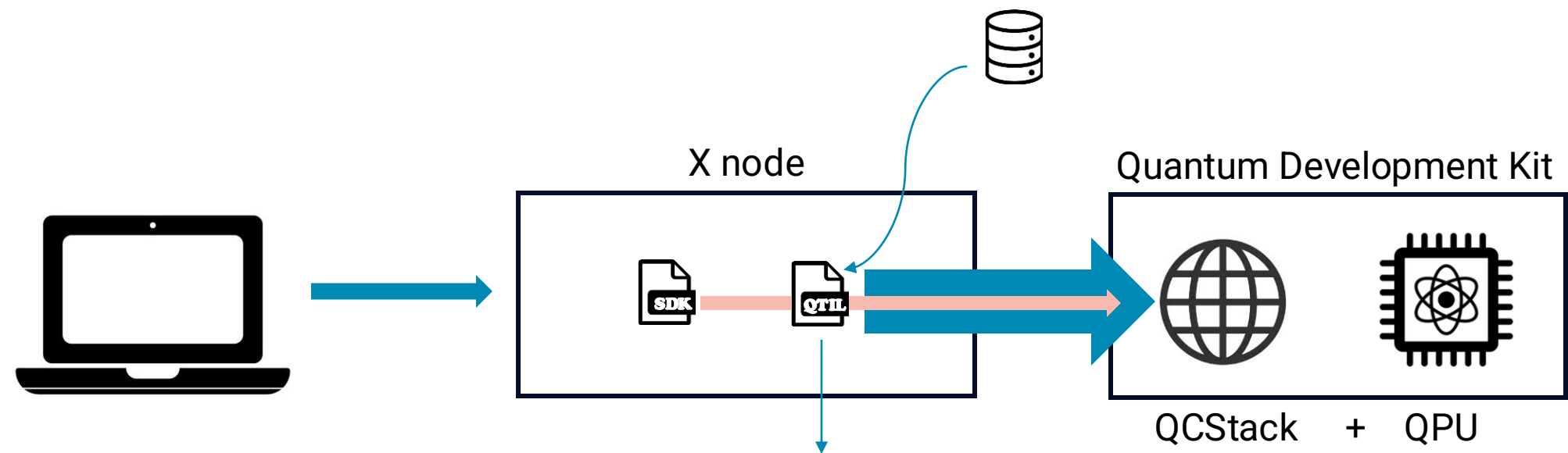
- Bus level communication (PCIe, NVLink, CXL)
- Molecular dynamics, ML applications
- Classical compute infrastructure

Quantum Queue Utility – QTIL

QCStack as a scheduler



QTIL reserves the QDK

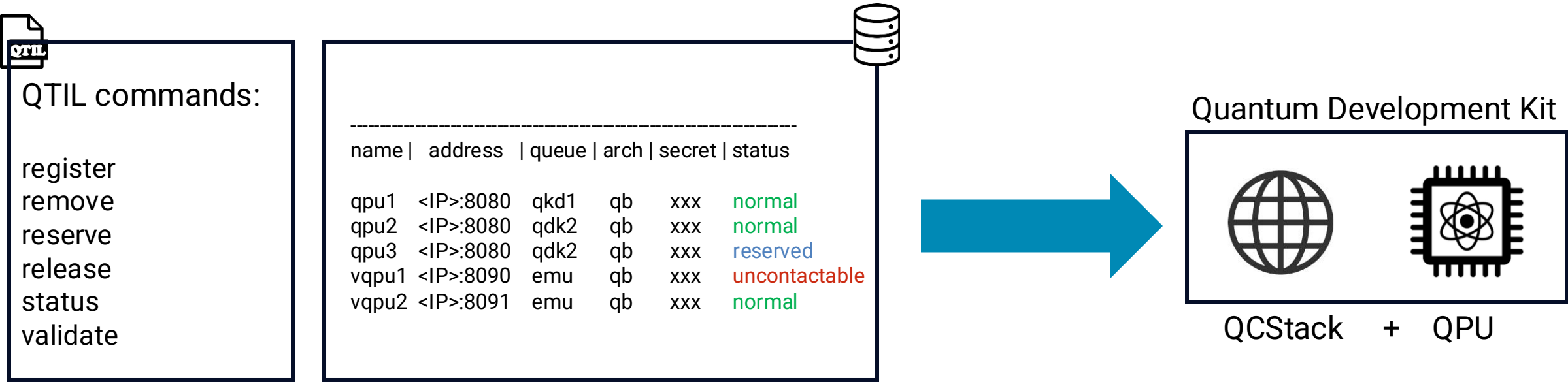


Client SDK submission is accepted only through QTIL reservation

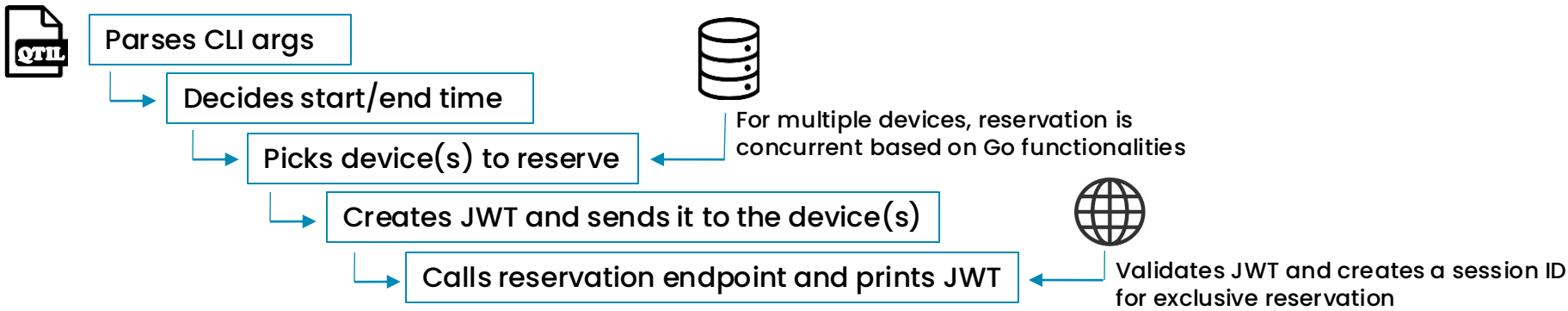
QTIL implements a QMRI (Quantum Resource Management Interface)

How QTIL works?

- 1 Register devices in the database with their own API secret keys (admin)

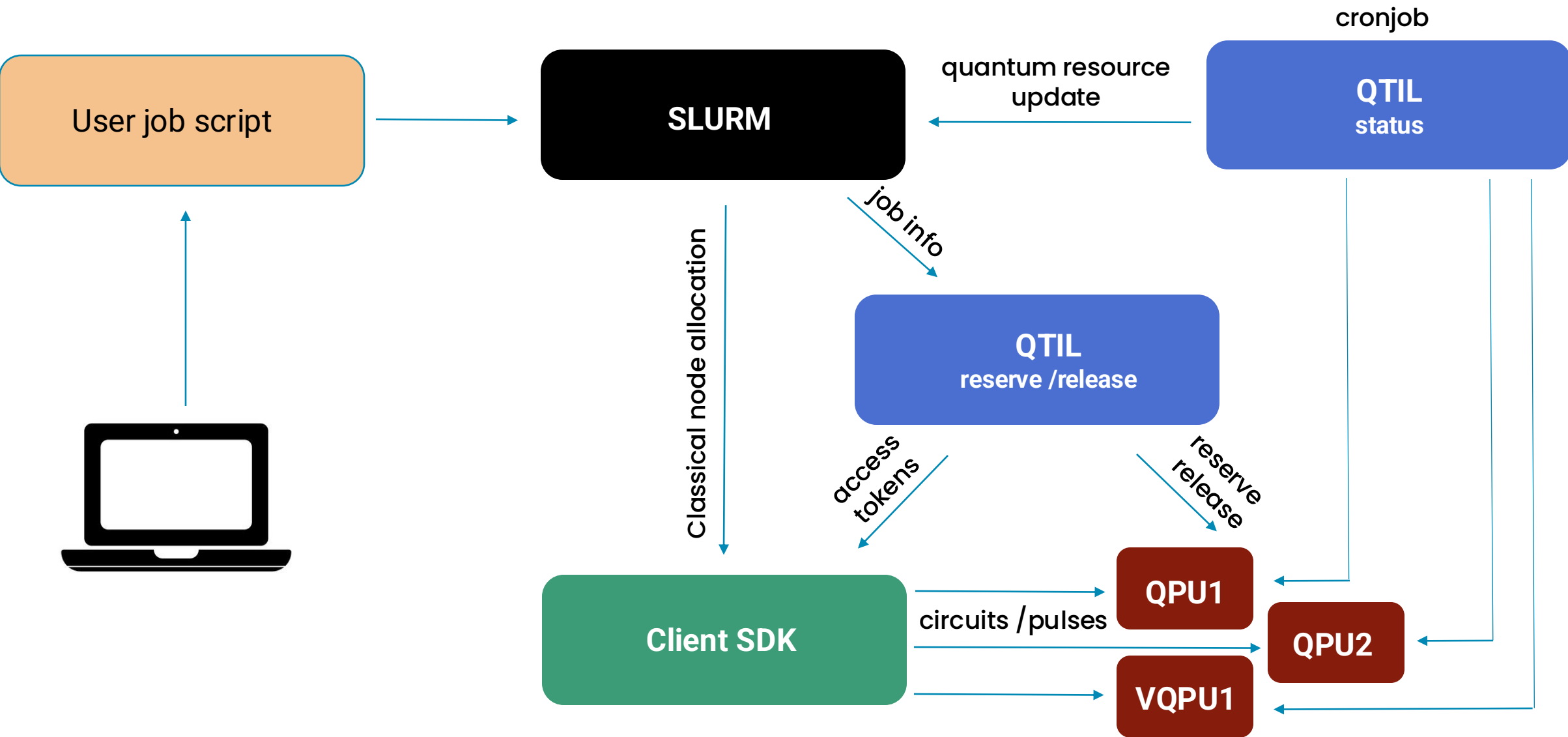


- 2 Reserve device, e.g., **qtil reserve -q qdk2 -d 60** -> only qpu2 and qpu3 could be reserved, if available, for 60 s



QTIL with SLURM

QTIL + SLURM in a nutshell



Thanks!